

Thin Cloud Removal Method Based on Low-Frequency Residual Diffusion and High-Frequency Modulation Refinement by Laplacian Pyramid Decoupling

Here, the cloud image is symbolized as I_{cloud} and the clear image is symbolized as I_{clear} . Then $I_{map} = I_{cloud} - I_{clear}$ is the residual of the two, which represents the directional degradation from the clear image to the cloud image.

① Forward Diffusion

In order to simulate the degradation process of cloud images, consider defining the forward process steps at time t from the cloud residual I_{map} :

$$I_t = I_{t-1} + I_{map}^t, \quad I_{map}^t \sim \mathcal{N}(\alpha_t I_{map}, \beta_t^2 \mathbf{I}) \quad (1)$$

Where I_{map}^t can be understood as the perturbation from I_{t-1} to I_t in the forward state. Since the degradation of cloud images is also affected by other atmospheric conditions, this paper defines I_{map}^t as consisting of cloud residuals and random noise, with α and β being independent scheduling coefficients that control them respectively. Here, the definition of formula (1) determines the initial state of the inverse process to perform denoising from impure noise. Furthermore, formula (1) is expanded to obtain the forward diffusion as:

$$\begin{aligned} I_t &= I_{t-1} + \alpha_t I_{map} + \beta_t \epsilon_{t-1} \\ &= I_{t-2} + (\alpha_{t-1} + \alpha_t) I_{map} + (\sqrt{\beta_{t-1}^2 + \beta_t^2}) \epsilon_{t-1} \\ &= \dots \\ &= I_{clear} + \bar{\alpha}_t I_{map} + \bar{\beta}_t \epsilon \end{aligned} \quad (2)$$

For the forward process from I_t to I_{t-1} , the transition probability matrix based on Bayes' theorem can be expressed as:

$$q(I_{t-1} | I_t, I_0, I_{map}) = q(I_t | I_{t-1}, I_0, I_{map}) \frac{q(I_{t-1} | I_0, I_{map})}{q(I_t | I_0, I_{map})} \quad (3)$$

Where the marginal probability distribution of the relevant forward process can be defined as:

$$q(I_t | I_{t-1}, I_0, I_{map}) = \mathcal{N}(I_t; I_{t-1} + \alpha_t I_{map}, \bar{\beta}_t^2 \mathbf{I}) \quad (4)$$

$$q(I_t | I_{clear}, I_{map}) = \mathcal{N}(I_t; I_{clear} + \bar{\alpha}_t I_{map}, \bar{\beta}_t^2 \mathbf{I}) \quad (5)$$

Where $\epsilon_{t-1}, \dots, \epsilon \in \mathcal{N}(0, \mathbf{I})$, $\bar{\alpha}_t = \sum_{i=1}^t \alpha_i$, $\bar{\beta}_t = \sqrt{\sum_{i=1}^t \beta_i^2}$. Therefore, the mean and variance are represented as follows:

$$\mu_t(I_t, I_{clear}, I_{map}) = I_t - \alpha_t I_{map} - \frac{\beta_t^2}{\bar{\beta}_t} \epsilon \quad (6)$$

$$\sum_t (I_t, I_{clear}, I_{map}) = \frac{\beta_t^2 \bar{\beta}_{t-1}^2}{\bar{\beta}_t^2} \quad (7)$$

Finally, the joint probability distribution during forward diffusion can be defined as:

$$q(I_{1:T} | I_{clear}, I_{map}) = \prod_{t=1}^T q(I_t | I_{t-1}, I_{map}) \quad (8)$$

$$q(I_t | I_{t-1}, I_{map}) = \mathcal{N}(I_t; I_{t-1} + \alpha_t I_{map}, \beta_t^2 \mathbf{I}) \quad (9)$$

② Inverse denoising

In the reverse process, I_{map} and ϵ are unknown quantities, and this paper trains a U-Net for prediction respectively.

According to formula (2), the target image is:

$$I_{clear}^\theta = I_t - \bar{\alpha}_t I_{map}^\theta - \bar{\beta}_t \epsilon_\theta \quad (10)$$

In the formula, θ represents the quantity that needs to be estimated using the model. Formula (10) once again proves the calculation method of reverse denoising of non-pure noise. The reverse process can be defined as:

$$p_\theta(I_{t-1} | I_t) = q_\sigma(I_{t-1} | I_t, I_{clear}^\theta, I_{map}^\theta) \quad (11)$$

Among them, q_σ is the transition probability from t to $t-1$, expressed as:

$$q_\sigma(I_{t-1} | I_t, I_{clear}, I_{map}) = \mathcal{N}(I_{t-1}; I_{clear} + \bar{\alpha}_{t-1} I_{map} + \frac{I_t - (I_{clear} + \bar{\alpha}_t I_{map})}{\beta_t}, \sigma_t^2 \mathbf{I}) \quad (12)$$

$$q(I_{t-1} | I_{clear}, I_{map}) = \mathcal{N}(\tilde{\mu}_{t-1}, \tilde{\Sigma}_{t-1}) \quad (13)$$

Where the mean and variance at time $t - 1$ can be expressed as:

$$\tilde{\mu}_{t-1} = I_0 + \bar{\alpha}_{t-1} I_{res} \quad (14)$$

$$\tilde{\Sigma}_{t-1} = \sigma_t^2 \mathbf{I} + (\frac{\sqrt{\bar{\beta}_{t-1}^2 - \sigma_t^2}}{\bar{\beta}_t})^2 \bar{\beta}_t^2 \mathbf{I} = \beta_{t-1}^2 \mathbf{I} \quad (15)$$

Combining formulas (13)(14)(15), formula (14) can be further expressed as:

$$q(I_{t-1} | I_0, I_{map}) = \mathcal{N}(I_{t-1}; I_0 + \bar{\alpha}_{t-1} I_{map}, \bar{\beta}_{t-1}^2 \mathbf{I}) \quad (16)$$

It can be seen from formula (17) that formula (13) can be extended to any time t . Therefore, the sampling process of inverse denoising is:

$$I_{t-1} = I_t - (\bar{\alpha}_t - \bar{\alpha}_{t-1}) I_{map}^\theta - (\bar{\beta}_t - \bar{\beta}_{t-1}) \epsilon_\theta \quad (17)$$